

Research Monographaskdfjhjk

How to Use the Book Template

Book Author

Month Year

Mục lục

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Preface

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part I.

First half of the book

chapter 1.

Introduction

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1.1. Research question

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1.2. Answer to the question

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1.3. Positioning in the literature

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chapter 2.

Sections and content

This chapter shows how to introduce sections, subsections, and paragraphs.¹ It also shows how to introduce various references into the book.

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2.1. Generic section

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¹A short footnote. The numbering of the footnotes restarts at the beginning of each chapter. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

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2.1.2. Another generic subsection

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2.2. Paragraphs

Here's the example of a run-in paragraph heading (*italic, on same line as text*).

Paragraph heading. Aenean fermentum purus id lacus volutpat, a eleifend mi posuere! Mauris nec nunc commodo, vehicula enim nec; vehicula ex. Nullam euismod lorem at eros efficitur: ut ultricies ante fringilla? Nam sagittis sapien id tortor commodo—a pulvinar velit ultricies... Integer ac magna velorci mollis vestibulum. Fusce id ipsum vel magna placerat vehicula. Curabitur ac lobortis justo.

Another paragraph with some numbers and special characters. Pellentesque habitant 25% morbi tristique senectus 1837–1905 et netus et malesuada fames ac turpis egestas. Integer semper euismod sapien vel dictum #1 and #6. Vivamus nec nunc sed metus interdum suscipit. (Maecenas tristique felis sed eleifend aliquet.) Donec et ipsum 3/4 in mauris ultricies pulvinar 9/2. Nullam quis “sapien a justo” vestibulum fermentum. Cras sed odio & vitae mi placerat mollis: \$23.

2.3. References

References can be appended at the end of a sentence, in parenthesis (?).² References can also include page numbers or other bibliographical information (?, p. 1305). References can be in text: for instance, ? found this and ?, figure 1 found that. It's also possible to collate several references to the same author: for instance, ?? found things. References from the same authors and same year appear as follows: ?? wrote something. It is also possible to cite the authors of a study by name, or the year of a study: for instance, ? wrote a paper in ? on this topic.

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2.4. URL

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chapter 3.

Math

This chapter displays a number of mathematical expressions to showcase the math fonts used in the template. The chapter also shows how to introduce different types of equations, theorems, propositions, corrolaries, lemmas, and other results into the book.

3.1. Roman letters

The Roman characters in math are just the same as the characters in the text—but in italic. Here are some small letters:

$$a\{p-r \times l\} + \frac{w(g/z+j)}{i(t)+j(t)+k(t)-e^p-x^j} + \frac{h[f]+x^f}{k[y]-e^y+x^y} = f(j)^{6+y} - i_{g,j,p}^3 \approx (p_{ji})^5.$$

Here are some capital letters as well:

$$G[p+P^7-Q]-A_B+L\{j\}=F(X)\rightarrow[Y+K]\times Z_f/H-[g_4-i], \text{ for any } i.$$

The punctuation is also the same in math as in text.¹

¹A longer footnote illustrating how multiple lines of footnote text affect the bottom of the page and the main text flow. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aliquam erat volutpat. Donec tincidunt, quam sed pellentesque fermentum, lorem dolor consequat lectus, at pulvinar arcu ipsum in ligula. Mauris pretium ipsum id sapien posuere vehicula. Vivamus consectetur, turpis vitae pellentesque molestie, libero odio venenatis mauris, id vestibulum nisl nulla sit amet enim. Cras ultricies consectetur ex, sit amet vestibulum lacus hendrerit vel. Sed lacinia eleifend purus. Vivamus consectetur, turpis vitae pellentesque molestie, libero odio venenatis mauris, id vestibulum nisl nulla sit amet enim. Cras ultricies consectetur ex, sit amet vestibulum lacus hendrerit vel. Sed lacinia eleifend purus.

3.2. Digits

The digits in math are also taken from the text font, just like the Roman characters. So 1, 2, 3, 4, 89, 03 (in text mode) is just the same as 1, 2, 3, 4, 89, 03 (in math mode). The percentage sign is also the same so 27% (in text mode) is the same as 27% (in math mode). Nulla facilisi.

3.3. Greek letters

Here are some small Greek letters in a math display:

$$\alpha^\theta + \gamma^4 + g(a) - \zeta \times \frac{y(\lambda)}{\kappa \cdot k + \sigma - s} \rightarrow \nu_\eta = \beta^\epsilon \cdot \delta - \mu + \frac{\xi^4}{\zeta_{ij}} \rightarrow m(x) + b^e + \omega.$$

Here are some capital Greek letters in another math display:

$$(3.1) \quad F(\Psi) - G(\Phi) \times \Delta^{10} = \Sigma \cdot \Omega^2 - \frac{\Lambda_i}{\Theta_j} \cdot \frac{\Gamma(x)}{\Pi(t)}.$$

Some but not all the variants of the standard Greek letters are available. There is a variant for theta ($\vartheta \neq \theta$), for epsilon ($\varepsilon \neq \epsilon$), for pi ($\varpi \neq \pi$), and for phi ($\varphi \neq \phi$). But there is no available variant for sigma ($\sigma = \sigma$), rho ($\rho = \rho$), and kappa (`\varkappa` gives an error).

3.4. Calligraphic letters

Calligraphic letters are taken from the same family of fonts as the Greek letters. The calligraphic letters are only available for capital letters. Here are some of the calligraphic letters:

$$(3.2) \quad \frac{\mathcal{A}(A)}{\mathcal{H}} - \mathcal{B}(\beta) + \mathcal{Q}(j + 3f) \rightarrow \mathcal{Y} + \mathcal{M}(\mathcal{N} - 2) = \mathcal{J}(X) \neq \mathcal{X}(J) - \mathcal{F}[\mathcal{D} \times \Delta]^8.$$

3.5. Blackboard-bold letters

Here are some blackboard-bold letters, used in some mathematical expressions: $x \in \mathbb{R}^n$ but $y \in \mathbb{Q}$ and $[A_1, A_2] \in \mathbb{Z} \times \mathbb{N}$; so that $\mathbb{E}[X \mid Y] = Y + 3 \cdot \mathbb{P}(X^2) \notin \mathbb{C} \times \mathbb{1}(X \subseteq \mathbb{B})$.

3.6. Symbols

The template comes with a full set of mathematical symbols. Here is a random assortment of some symbols:

$$(3.3) \quad \left(\frac{\sum_i \tilde{z}_i}{\sum_{j=-\infty}^{+\infty} X_j} \right) \pm Q \subset \mathbb{R} \neq \mathbb{C} \implies \sqrt{\frac{\mathcal{X}}{2}} > \sqrt{\mathcal{Y}} \gg \hat{b}(y_0).$$

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$$(3.4) \quad \sum_i \left[\frac{\hat{x}(i)}{\mathcal{Y}(i)} \right]^2 \leq 2\pi \Rightarrow \prod_j \bar{y}(j)^2 \geq \left\{ \iint \frac{g(h)}{\gamma(s) \times \Delta(h)} dh ds \right\} \mapsto f \circ g(x) \iff \tilde{Q}(x) \propto \frac{\dot{x}}{2}.$$

3.7. Bold characters

In the template it is possible to bold all math characters. Roman characters can be bolded, such as **$\mathbf{a} + \mathbf{D} = \mathbf{E}^2 + \mathbf{j}/\mathbf{i}$** . Greek characters can also be bolded, such as **$\boldsymbol{\alpha} + \boldsymbol{\Delta} = \boldsymbol{\epsilon}^2 + \boldsymbol{\Lambda}/\boldsymbol{\Phi}$** . It is also possible to bold digits: **$1 + 2 \neq \mathbf{1} + \mathbf{2}$** . Finally, it is possible to bold calligraphic letters: **$\mathcal{C} + \mathcal{E} - [\mathcal{X} + \mathcal{Y}]$** . Blackboard-bold letters are already “bold” so they cannot be bolded further.

3.8. Operators

Operators in mathematical expressions are typeset with the text font. Here are a few examples:

$$(3.5) \quad \max_x \pi(x) = \frac{\int \ln(u) du}{\int \exp(v) dv} \mapsto \cos(\theta) + \sin(\omega) - \min_z [\tan(z)].$$

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3.9. Some equations

This subsection illustrates how the different math fonts fit together. The equations below combine various characters together.

First, here are equations in text. Lorem ipsum dolor sit amet, consectetur adipiscing elit: $\prod_i \alpha_i(X) \geq 2 \cdot [\partial a_u(Z)/\partial Z] \propto \mathcal{X} \sim 5.2\pi \rightsquigarrow +\infty$. Lorem ipsum dolor sit amet, consectetur adipiscing elit: $\sum_{x=1}^4 \xi^x(j) = 0, \forall j \in \mathbb{R}$. Here are some other expressions: $\mathcal{R}(z) = \max\{T_1, \dots, T_n\} > \hat{z}$, and $\mathbb{P}(T_n < z) = 1 - S(z) = F(z)$. Sed vehicula ultrices odio, vitae maximus justo convallis at. Fusce quis consequat arcu: $\alpha^* = S(z^*)$ and $S^*(z^*) = \mathbb{E}[\alpha \mid \beta]$. Integer non ante eget purus aliquet consequat. Mauris ut dui nec ligula consectetur commodo. Mauris ut dui nec ligula consectetur commodo. Mauris ut dui nec ligula consectetur commodo. Mauris ut dui nec ligula consectetur commodo—mauris ut dui nec ligula consectetur commodo.

Here are equations in display mode, without labels:

$$\frac{\mathcal{S}(u^*)}{1 - \frac{\partial \ln(\theta)}{\partial \theta} + [\gamma^\eta] - \mathcal{G}(z^*)} = \sum_{i=0}^{+\infty} \frac{dF^i(x)}{dx} - \mathcal{A}(i) \geq \left\{ \sum_{i=0}^{+\infty} G^i - \Omega(i) \right\}$$

$$\frac{\mathbb{Q}(z^*)}{1 - \mu + \exp(\xi) \times \mathbb{M}(z^*)} = \prod_{i=0}^{+\infty} F^i - \Phi(i) \neq \left(\int_0^{+\infty} \frac{G^i}{\epsilon(i)} di \right).$$

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vehicula ultrices odio, vitae maximus justo convallis at. Fusce quis consequat arcu. Integer non ante eget purus aliquet consequat. Mauris ut dui nec ligula consectetur commodo.

And here is a single equation in display mode without a label:

$$\dot{y}(t) \rightarrow \sum_{j=0}^{+\infty} \mathcal{W}^j(z^*(t)) = \frac{S(z)\mathbb{E}(N(z))}{\mu_1 - \zeta_2} \cdot 2 - \left[\prod_{i=0}^{+\infty} \mathcal{F}^i \right] - \iiint \exp(\lambda(t)\mu(s)) \sin(\theta) dt ds d\theta.$$

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam consequat, ipsum eget tincidunt aliquam, mi est pellentesque lacus, nec ullamcorper leo magna a dui.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aliquam erat volutpat. Donec tincidunt, quam sed pellentesque fermentum, lorem dolor consequat lectus, at pulvinar arcu ipsum in ligula. Mauris pretium ipsum id sapien posuere vehicula. Vivamus consectetur, turpis vitae pellentesque molestie, libero odio venenatis mauris, id vestibulum nisl nulla sit amet enim. Cras ultricies consectetur ex, sit amet vestibulum lacus hendrerit vel. Sed lacinia eleifend purus. Nunc dignissim eros et sapien lacinia eleifend. Curabitur sit amet velit eu metus suscipit tempus. Maecenas in libero felis. Sed quis eros sapien. Suspendisse potenti. Curabitur porta est sed enim efficitur, a tempor justo fermentum. Vivamus pulvinar enim vitae semper rutrum. Aenean non mauris sed lectus vestibulum feugiat eget et ipsum. Donec semper vitae enim at ultricies. Sed eu arcu justo.

3.10. Some propositions, theorems, and lemmas

Here is a proposition with some more math. Note that numbering of all the results is specific to each chapter:

proposition 3.1. Lorem ipsum dolor sit amet, consectetur adipiscing elit:

$$(3.6) \quad \sum_k \mathbf{s}_{k_x}(z) \approx \frac{S(z)^x}{k/23 - \zeta\gamma[45 - S(z)] + \ln(y) - j^2 + x(l)}.$$

Ut enim ad minim veniam, sunt in culpa qui officia deserunt mollit anim id est laborum. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Chúng minh. Here is the proof to the proposition. Donec commodo justo a eros malesuada, eget vulputate tortor accumsan. Sed ac pulvinar nulla. Etiam quis felis dapibus, vulputate metus eu, finibus nunc. Sed vel sodales dui. Nam venenatis dolor non orci tempus fermentum. Vivamus sodales justo a ligula cursus aliquet. Sed fringilla nunc vitae justo finibus, id placerat lectus sodales. $\square$

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Nullam varius nisi vel ipsum vehicula, at finibus justo ultrices. Morbi vitae leo vitae purus dapibus tempor. Vestibulum sed nisi sit amet mi suscipit suscipit non nec tortor. Vivamus non ligula id turpis tempus fermentum ac sit amet leo. Etiam sit amet magna at turpis fringilla vestibulum. Sed non nisl eu velit consectetur malesuada eget eu libero. Sed a nisi sit amet velit facilisis feugiat nec at est. Quisque nec ultrices purus. Maecenas gravida mauris vel nulla eleifend, non dictum enim ullamcorper. Sed sit amet turpis in lectus dictum facilisis. Donec sed eros sed dui consectetur dictum. Sed volutpat tortor nec odio viverra, id sodales enim congue. Maecenas sit amet eros vitae nisl placerat mattis. Integer vitae metus eget risus vulputate consectetur. Suspendisse potenti.

Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae;

Sed ac eros vel felis vehicula vehicula. Phasellus interdum justo a felis congue, vel dapibus est tristisque. Now here is a lemma—numbering is specific to each type of entry, so lemmas have a separate numbering from propositions:

lemma 3.1. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat:

$$(3.7) \quad z^* = \int_0^\infty \alpha(i) \cdot \frac{1 - \beta}{1 - \alpha(i)\beta} di.$$

Nullam varius nisi vel ipsum vehicula, at finibus justo ultrices. Morbi vitae leo vitae purus dapibus tempor: $\int_0^\infty \alpha(i) di < +\infty$.

Here is the proof to the lemma. Donec commodo justo a eros malesuada, eget vulputate tortor accumsan:

Chứng minh. Sed ac pulvinar nulla: $\alpha^* = S(z^*)^2 + \beta \times \gamma^2$ et $S^*(z^*) = \mathbb{E}[\alpha - \gamma \mid \beta] \in \mathbb{R}^+$. Etiam quis felis dapibus, vulputate metus eu, finibus nunc. Sed vel sodales dui. Nam venenatis dolor non orci tempus fermentum. Vivamus sodales justo a ligula cursus aliquet. Sed fringilla nunc vitae justo finibus, id placerat lectus sodales. $\square$

Quia dolor sit amet consectetur adipiscing velit, sed quia non numquam eius modi tempora incidunt, ut labore et dolore magnam aliquam quaerat voluptatem. Then here is another lemma:

lemma 3.2. Consectetur adipiscing elit, sed do eiusmod tempor incidunt ut labore et dolore magna aliqua $\mathcal{Z}(\alpha)$.

$$\frac{\sum_i z^i}{\prod_i q^i} \rightarrow \frac{\int_0^\infty \alpha(i) \cdot [1 - \beta] di}{1 - \exp(\alpha) \sin(\beta)}.$$

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur: $\alpha(i) \cdot [1 - \beta] < \zeta$. Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum.

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a tempor justo fermentum. Vivamus pulvinar enim vitae semper rutrum. Aenean non mauris sed lectus vestibulum feugiat eget et ipsum. Donec semper vitae enim at ultricies. Sed eu arcu justo. And here is a theorem following the lemma:

theorem 3.1. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua:

$$\mathbb{E}(N(z^*)) \approx \frac{1 - \mathbb{P}(\alpha\pi)}{1 - \pi} - \frac{f(y)}{z(p)^*} + P(\Gamma).$$

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum.

Nullam varius nisi vel ipsum vehicula, at finibus justo ultrices. Morbi vitae leo vitae purus dapibus tempor. Vestibulum sed nisi sit amet mi suscipit suscipit non nec tortor. Vivamus non ligula id turpis tempus fermentum ac sit amet leo. Etiam sit amet magna at turpis fringilla vestibulum. Sed non nisl eu velit consectetur malesuada eget eu libero. Sed a nisi sit amet velit facilisis feugiat nec at est. Quisque nec ultrices purus. Maecenas gravida mauris vel nulla eleifend, non dictum enim ullamcorper. Sed sit amet turpis in lectus dictum facilisis. Donec sed eros sed dui consectetur dictum. Sed volutpat tortor nec odio viverra, id sodales enim congue. Maecenas sit amet eros vitae nisl placerat mattis. Integer vitae metus eget risus vulputate consectetur. Suspendisse potenti.

part II.

Second half of the book

chapter 4.

Thiết kế vật lý

4.1. Script DDL PostgreSQL cho Lược đồ Cơ sở dữ liệu

```
1
2 CREATE TABLE templates (
3     template_id        UUID           NOT NULL DEFAULT gen_random_uuid() ,
4     workspace_id        UUID           NOT NULL,
5     name                VARCHAR(150)   NOT NULL,
6     description         TEXT           NULL,
7     content_structure    TEXT           NULL,
8     creator_type        VARCHAR(20)    NOT NULL,
9     status              VARCHAR(20)    NOT NULL DEFAULT 'DRAFT' ,
10    created_at           TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
11    updated_at           TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
12    CONSTRAINT pk_templates PRIMARY KEY (template_id),
13
14    CONSTRAINT uq_templates_workspace_name UNIQUE (workspace_id, name),
15    CONSTRAINT fk_templates_workspace
16        FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
17        ON UPDATE CASCADE
18        ON DELETE CASCADE,
19    CONSTRAINT ck_templates_creator_type
20        CHECK (creator_type IN ('SYSTEM', 'USER', 'AI')),
21    CONSTRAINT ck_templates_status
22        CHECK (status IN ('DRAFT', 'ACTIVE', 'ARCHIVED')),
23    CONSTRAINT ck_templates_name_not_blank
24        CHECK (length(btrim(name)) > 0),
```

```

25     CONSTRAINT ck_templates_updated_after_created
26         CHECK (updated_at >= created_at)
27 );
28
29 CREATE TABLE template_sections (
30     section_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
31     template_id         UUID          NOT NULL,
32     heading_title       VARCHAR(255)  NOT NULL,
33     order_index         INTEGER       NOT NULL,
34     prompt_instruction  TEXT          NULL,
35     created_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
36     updated_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
37     CONSTRAINT pk_template_sections PRIMARY KEY (section_id) ,
38
39     CONSTRAINT uq_template_sections_order UNIQUE (template_id, order_index) ,
40     CONSTRAINT fk_template_sections_template
41         FOREIGN KEY (template_id) REFERENCES templates (template_id)
42         ON UPDATE CASCADE
43         ON DELETE CASCADE,
44     CONSTRAINT ck_template_sections_order_index
45         CHECK (order_index >= 0) ,
46     CONSTRAINT ck_template_sections_heading_not_blank
47         CHECK (length(btrim(heading_title)) > 0)
48 );
49
50 CREATE TABLE placeholders (
51     placeholder_id      UUID          NOT NULL DEFAULT gen_random_uuid() ,
52     section_id          UUID          NOT NULL,
53     variable_name       VARCHAR(100)  NOT NULL,
54     expected_data_type  VARCHAR(30)   NOT NULL,
55     description         TEXT          NULL,
56     is_required         BOOLEAN       NOT NULL DEFAULT TRUE,
57     created_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
58     updated_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
59     CONSTRAINT pk_placeholders PRIMARY KEY (placeholder_id) ,
60
61     CONSTRAINT uq_placeholders_section_variable UNIQUE (section_id ,
62     variable_name) ,
63     CONSTRAINT fk_placeholders_section
64         FOREIGN KEY (section_id) REFERENCES template_sections (section_id)
65         ON UPDATE CASCADE
66         ON DELETE CASCADE,
67     CONSTRAINT ck_placeholders_expected_data_type
68         CHECK (expected_data_type IN ( 'STRING' , 'NUMBER' , 'BOOLEAN' ,
69         'DATE' , 'LIST' , 'OBJECT' , 'CODE' ) ) ,
70
71     CONSTRAINT ck_placeholders_variable_name_format

```

```

71         CHECK (variable_name ~ '^[A-Za-z_][A-Za-z0-9_]*$')
72     );
73
74 CREATE TABLE template_mappings (
75     mapping_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
76     placeholder_id      UUID          NOT NULL,
77     code_entity_id      UUID          NOT NULL,
78     extraction_rule     TEXT          NULL,
79     is_active           BOOLEAN       NOT NULL DEFAULT TRUE,
80     CONSTRAINT pk_template_mappings PRIMARY KEY (mapping_id) ,
81
82     CONSTRAINT uq_template_mappings_pair UNIQUE (placeholder_id ,
83     code_entity_id) ,
84     CONSTRAINT fk_template_mappings_placeholder
85         FOREIGN KEY (placeholder_id) REFERENCES placeholders (placeholder_id
86     )
87         ON UPDATE CASCADE
88         ON DELETE CASCADE,
89     CONSTRAINT fk_template_mappings_code_entity
90         FOREIGN KEY (code_entity_id) REFERENCES code_entities (
91     code_entity_id)
92         ON UPDATE CASCADE
93         ON DELETE CASCADE
94 );
95
96 CREATE TABLE documents (
97     document_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
98     workspace_id         UUID          NOT NULL,
99     author_user_id       UUID          NOT NULL,
100    title                 VARCHAR(255) NOT NULL,
101    document_type         VARCHAR(30)  NOT NULL,
102    status                VARCHAR(20)  NOT NULL DEFAULT 'DRAFT' ,
103    created_at            TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
104    updated_at            TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
105    deleted_at            TIMESTAMP     NULL,
106    CONSTRAINT pk_documents PRIMARY KEY (document_id) ,
107    CONSTRAINT fk_documents_workspace
108        FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
109        ON UPDATE CASCADE
110        ON DELETE CASCADE,
111    CONSTRAINT fk_documents_author
112        FOREIGN KEY (author_user_id) REFERENCES users (user_id)
113        ON UPDATE CASCADE
114        ON DELETE RESTRICT,
115    CONSTRAINT ck_documents_type
116        CHECK (document_type IN ( 'SRS' , 'API_DOC' , 'USER_GUIDE' ,
117        'RELEASE_NOTE' , 'TECH_DESIGN' , 'OTHER' ) ) ,

```

```

115     CONSTRAINT ck_documents_status
116         CHECK (status IN ('DRAFT', 'IN_REVIEW', 'PUBLISHED', 'ARCHIVED')),
117     CONSTRAINT ck_documents_title_not_blank
118         CHECK (length(btrim(title)) > 0),
119     CONSTRAINT ck_documents_deleted_after_created
120         CHECK (deleted_at IS NULL OR deleted_at >= created_at)
121 );
122
123 CREATE UNIQUE INDEX uq_documents_workspace_title_active
124     ON documents (workspace_id, title)
125     WHERE deleted_at IS NULL;
126
127 CREATE TABLE document_versions (
128     version_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
129     document_id         UUID          NOT NULL,
130     version_number      VARCHAR(20)   NOT NULL,
131     content             TEXT          NULL,
132     change_summary      TEXT          NULL,
133     status              VARCHAR(20)   NOT NULL DEFAULT 'DRAFT',
134     ticket_codes        VARCHAR(500)  NULL,
135     created_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
136     updated_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
137     CONSTRAINT pk_document_versions PRIMARY KEY (version_id),
138
139     CONSTRAINT uq_document_versions_number UNIQUE (document_id,
140     version_number),
141     CONSTRAINT fk_document_versions_document
142         FOREIGN KEY (document_id) REFERENCES documents (document_id)
143         ON UPDATE CASCADE
144         ON DELETE CASCADE,
145     CONSTRAINT ck_document_versions_status
146         CHECK (status IN ('DRAFT', 'IN_REVIEW', 'APPROVED', 'REJECTED', '
147     PUBLISHED')),
148
149     CONSTRAINT ck_document_versions_number_format
150         CHECK (version_number ~ '^[vV]?[0-9]+(\.[0-9]+){0,2}$')
151 );
152
153 CREATE TABLE change_logs (
154     log_id              UUID          NOT NULL DEFAULT gen_random_uuid(),
155     version_id          UUID          NOT NULL,
156     change_type         VARCHAR(20)   NOT NULL,
157     change_description  TEXT          NULL,
158     diff_content        TEXT          NULL,
159     created_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
160     updated_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
161     CONSTRAINT pk_change_logs PRIMARY KEY (log_id),

```

```

160     CONSTRAINT fk_change_logs_version
161         FOREIGN KEY (version_id) REFERENCES document_versions (version_id)
162         ON UPDATE CASCADE
163         ON DELETE CASCADE,
164     CONSTRAINT ck_change_logs_change_type
165         CHECK (change_type IN ( 'CREATE', 'UPDATE', 'DELETE', 'RESTORE', '
PUBLISH' ))
166 );
167
168
169 CREATE INDEX idx_templates_workspace_id ON templates (workspace_id);
170 CREATE INDEX idx_templates_status ON templates (status);
171 -- placeholders / mappings
172 CREATE INDEX idx_template_mappings_code_entity ON template_mappings (
code_entity_id);
173 CREATE INDEX idx_template_mappings_active ON template_mappings (
placeholder_id)
174 WHERE is_active = TRUE;
175
176 CREATE INDEX idx_documents_workspace_id ON documents (workspace_id);
177 CREATE INDEX idx_documents_author ON documents (author_user_id);
178 CREATE INDEX idx_documents_status ON documents (status);
179 CREATE INDEX idx_documents_created_at ON documents (created_at DESC);
180
181 CREATE INDEX idx_document_versions_status ON document_versions (status);
182 CREATE INDEX idx_document_versions_created_at ON document_versions (
document_id, created_at DESC);
183
184 CREATE INDEX idx_change_logs_version_id ON change_logs (version_id);
185 CREATE INDEX idx_change_logs_created_at ON change_logs (created_at DESC);
186
187 CREATE OR REPLACE FUNCTION set_updated_at()
188 RETURNS TRIGGER AS $$
189 BEGIN
190     NEW.updated_at := CURRENT_TIMESTAMP;
191     RETURN NEW;
192 END;
193 $$ LANGUAGE plpgsql;
194 CREATE TRIGGER trg_templates_updated_at
195     BEFORE UPDATE ON templates
196     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
197 CREATE TRIGGER trg_template_sections_updated_at
198     BEFORE UPDATE ON template_sections
199     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
200 CREATE TRIGGER trg_placeholders_updated_at
201     BEFORE UPDATE ON placeholders
202     FOR EACH ROW EXECUTE FUNCTION set_updated_at();

```

```
203 CREATE TRIGGER trg_documents_updated_at
204     BEFORE UPDATE ON documents
205     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
206 CREATE TRIGGER trg_document_versions_updated_at
207     BEFORE UPDATE ON document_versions
208     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
209 CREATE TRIGGER trg_change_logs_updated_at
210     BEFORE UPDATE ON change_logs
211     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
```

STAGE 3 - DDL SCRIPT (PostgreSQL)